SaaS Revenue & Retention Analysis

SQL Portfolio Project

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Tools: **SQL (PostgreSQL / BigQuery), Excel, GitHub**

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1. Executive Summary

This project analyzes a SaaS-style dataset to evaluate revenue performance, acquisition channel effectiveness, and customer concentration.

The analysis reveals three key findings.

First, revenue is highly concentrated among a small number of customers. The top two customers contribute more than 65% of total revenue, creating potential risk if a major customer churns.

Second, paid marketing and outbound sales generate the majority of revenue. These channels appear to be the most effective acquisition strategies for the business.

Third, monthly revenue fluctuates significantly rather than growing steadily. This suggests that revenue may be influenced by campaign activity or large individual payments rather than stable recurring growth.

To achieve more sustainable growth, the company should focus on expanding the number of paying customers, improving retention, and reducing dependency on a few high-value accounts.

2. Project Overview

This project analyzes a SaaS-style dataset to understand revenue dynamics and customer value across the business.

The analysis focuses on several key questions commonly addressed by data analysts in SaaS companies:

- Which acquisition channels generate the most revenue?
- How does revenue evolve over time?
- Are revenues concentrated among a few customers?
- Which subscription plans generate the most MRR?
- How well are customer cohorts retained over time?
- Which channels bring higher-value customers?
- What does customer lifetime value look like across the business?

The dataset contains three core tables: customers, payments, and subscriptions. SQL queries are used to aggregate revenue metrics, evaluate customer contributions, and analyze subscription performance.

This project demonstrates practical SQL techniques used in business analysis:

- JOIN operations
- Aggregations using SUM, COUNT, and AVG
- Common Table Expressions (CTEs)
- Window functions
- Cohort retention analysis
- Revenue share and concentration analysis
- SaaS MRR metrics
- Customer lifetime value (LTV) analysis

The goal is to demonstrate how raw transactional data can be transformed into actionable business insights.

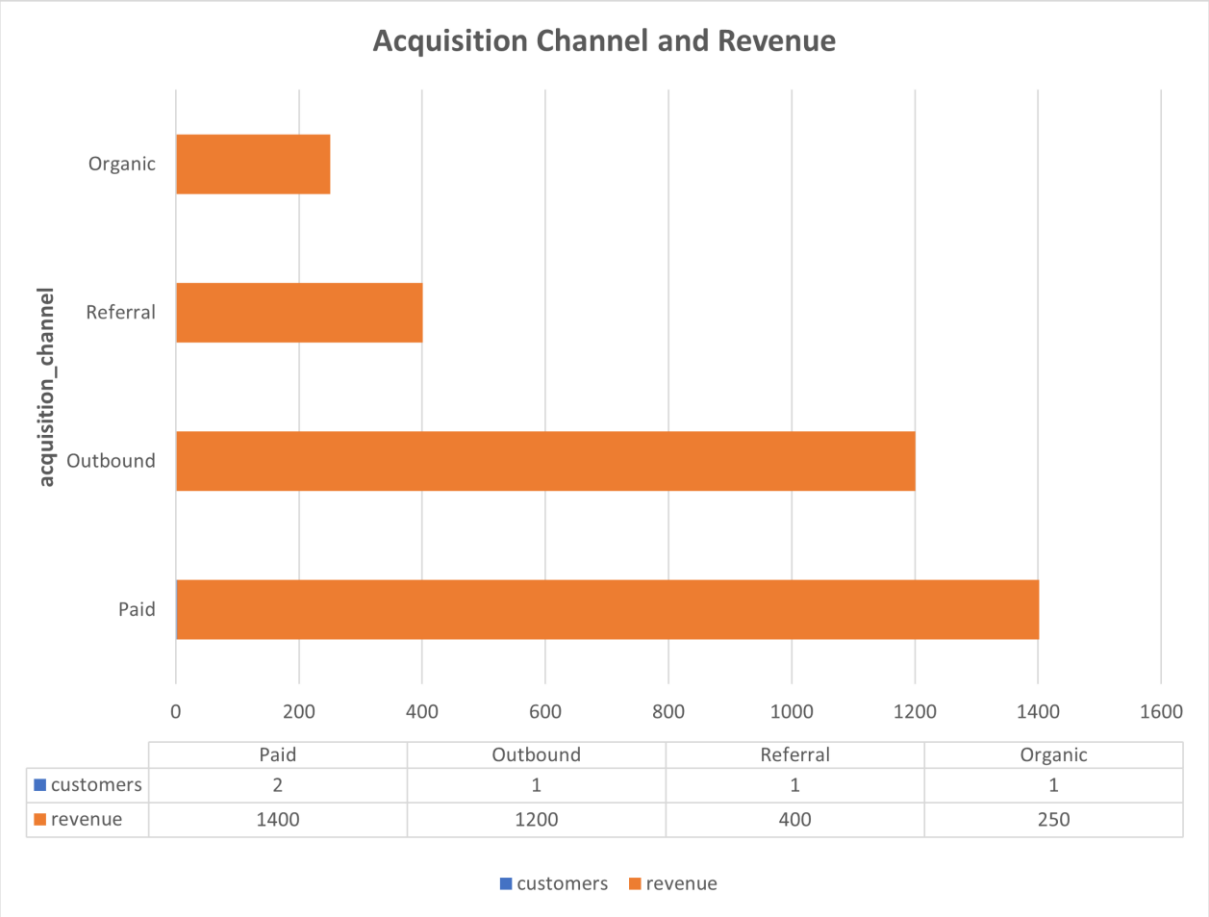
3. Key Metrics

Metric	Description
Monthly Revenue	Total revenue generated from successful payments per month
MRR (Monthly Recurring Revenue)	Revenue generated by active subscription plans
Revenue Share	Percentage contribution of a customer or channel to total revenue
Customer Lifetime Value (LTV)	Total revenue generated by a customer over time
Revenue per Customer	Average revenue generated per paying customer

These metrics represent core SaaS KPIs used to evaluate growth, retention, and monetization performance.

4. Analysis Findings

4.1 Revenue by Acquisition Channel



	Paid	Outbound	Referral	Organic
customers	2	1	1	1
revenue	1400	1200	400	250

Figure 1: Revenue by Acquisition Channel

Paid and Outbound channels generate the largest share of revenue.

Paid acquisition contributes the highest total revenue, indicating that marketing campaigns are a major growth driver. Outbound sales is the second strongest channel, suggesting that direct sales outreach also plays a significant role.

Referral and Organic channels contribute smaller portions of revenue but may represent lower acquisition costs.

4.2 Monthly Revenue Trend

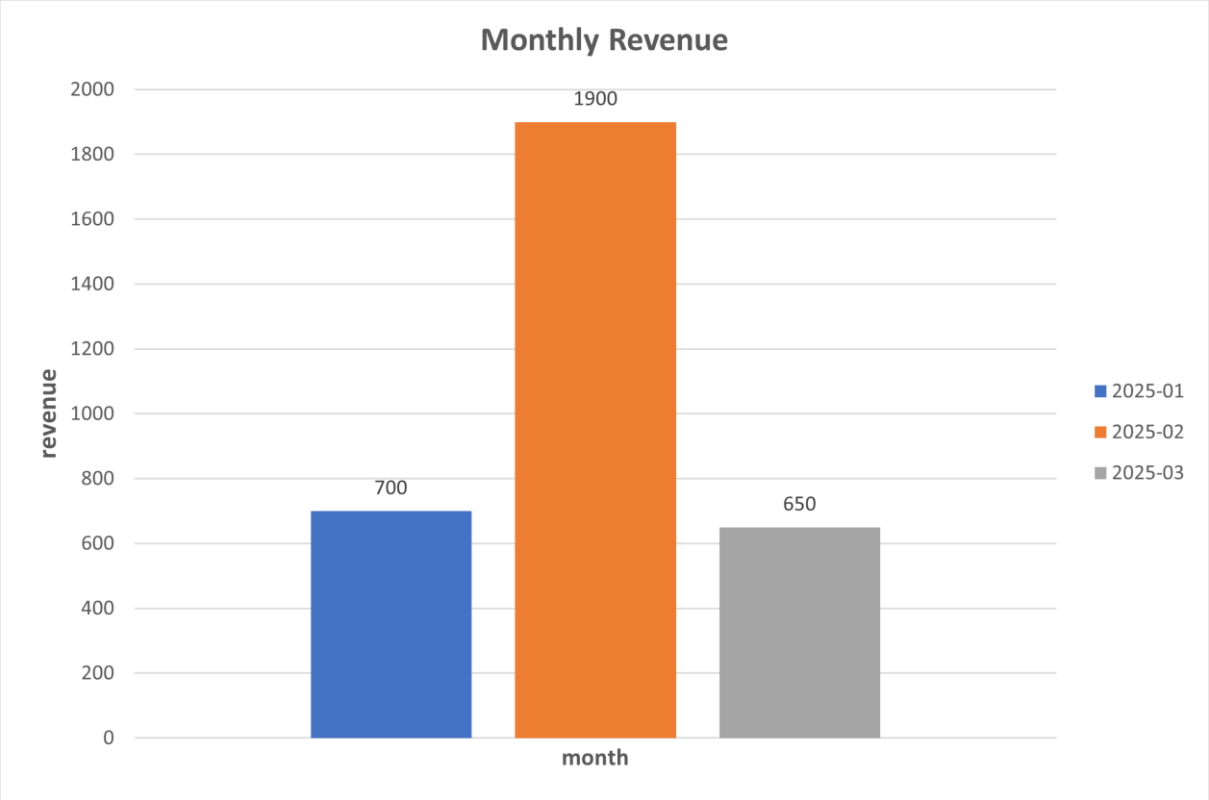


Figure 2: Monthly Revenue Trend

Monthly revenue fluctuates across the observed period.

Revenue peaks in February before declining in March. This pattern suggests that revenue may be influenced by short-term campaigns or individual large payments rather than stable recurring growth.

A mature SaaS business typically shows steady monthly growth driven by recurring subscriptions.

4.3 Customer Revenue Share

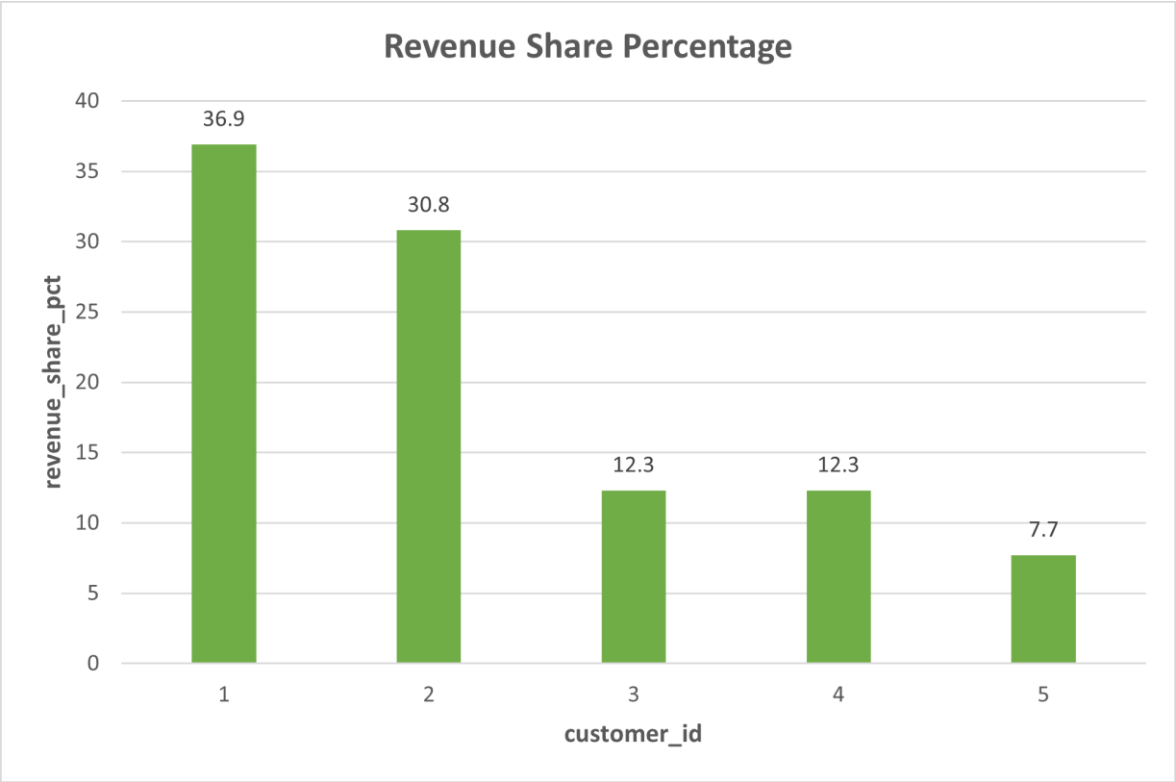


Figure 3: Customer Revenue Share

Revenue distribution across customers is highly uneven.

Customer 1 generates the largest share of revenue, followed by Customer 3 and Customer 2. The top two customers contribute more than 65% of total revenue.

This indicates strong revenue concentration and potential exposure if one of these customers churns.

4.4 MRR Distribution by Subscription Plan

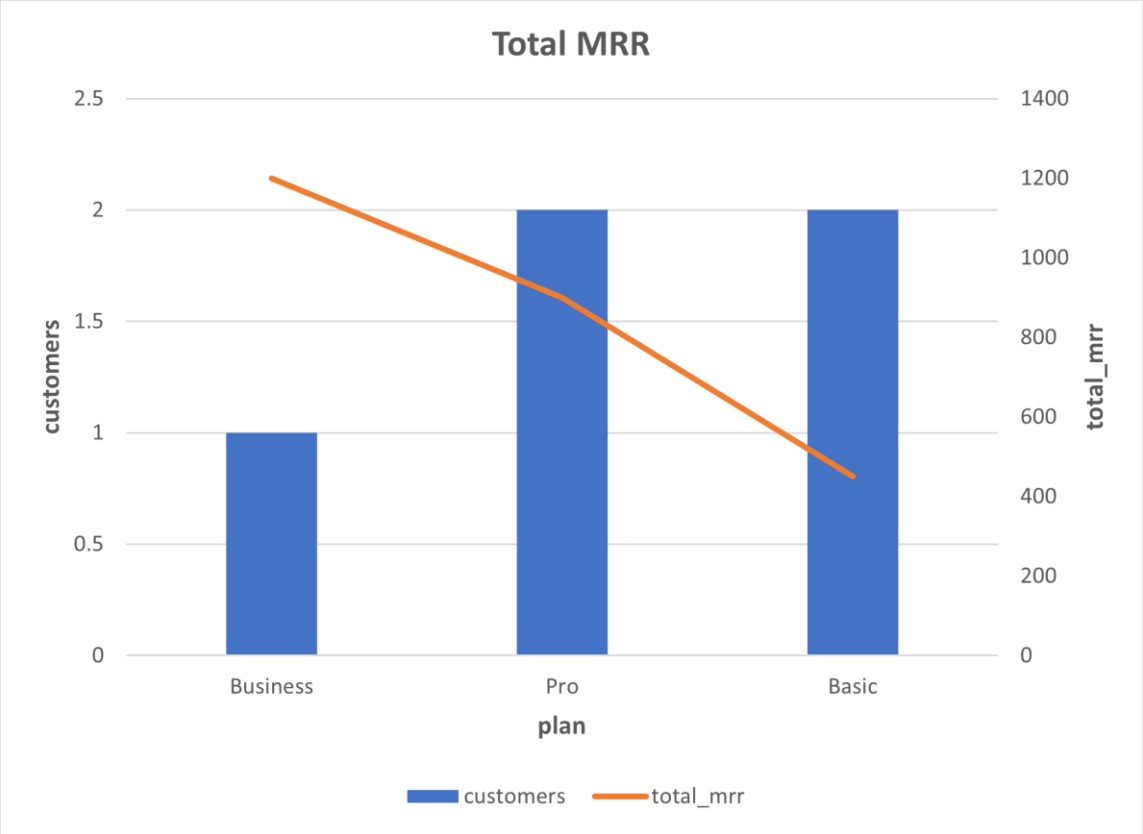


Figure 4: MRR Distribution by Subscription Plan

Subscription plans contribute differently to monthly recurring revenue.

Higher-tier plans generate the largest share of MRR despite having fewer customers. Lower-tier plans attract more users but generate less revenue per customer.

This suggests that upselling customers to higher-tier plans could significantly increase overall revenue.

5. Business Insights

5.1 Revenue Concentration Risk

Revenue is concentrated among a small number of customers. If one of the largest customers churns, the company could experience a significant drop in revenue.

Expanding the number of mid-sized customers would reduce this risk.

5.2 Customer Dependency

The business currently depends on several high-value accounts rather than a diversified customer base.

A more balanced revenue distribution would improve financial stability and reduce exposure to individual customer churn.

5.3 Growth Sustainability

Revenue growth appears to be driven more by individual payments than by broad customer growth.

Sustainable SaaS growth typically depends on:

- increasing the number of paying customers
- improving retention rates
- expanding average revenue per user

The data suggests that customer base expansion may be the most effective long-term growth strategy.

6. Methodology (SQL Workflow)

The analysis follows a structured SQL workflow.

Step 1: Data Exploration

The dataset includes three tables: customers, payments, and subscriptions.

Step 2: Revenue Aggregation

```
SELECT
DATE_TRUNC('month', payment_date) AS month,
SUM(amount_usd) AS revenue
FROM payments
WHERE status = 'paid'
GROUP BY DATE_TRUNC('month', payment_date)
ORDER BY month;
```

This query calculates total revenue generated each month.

Step 3: Channel Performance Analysis

```
SELECT
c.acquisition_channel,
COUNT(DISTINCT p.customer_id) AS paid_customers,
SUM(p.amount_usd) AS revenue
FROM payments p
JOIN customers c
ON p.customer_id = c.customer_id
WHERE p.status = 'paid'
GROUP BY c.acquisition_channel
ORDER BY revenue DESC;
```

This query evaluates which acquisition channels generate the most revenue.

Step 4: Customer Lifetime Value

```
SELECT
customer_id,
SUM(amount_usd) AS lifetime_value,
COUNT(*) AS payment_count
FROM payments
WHERE status = 'paid'
GROUP BY customer_id
ORDER BY lifetime_value DESC;
```

This query calculates total lifetime revenue generated by each customer.

GitHub Repository

Full project:

https://github.com/terrynguyen19881215-del/sql-sales-analysis